Foundation Programme
Universiti Teknologi Malaysia
Session 2021/2022 Semester 3

Test 1 (15%)

FPSM 0024 Calculus

Answer all SIX questions. Show all your work clearly. Time given is 1 hour and 15 minutes with additional 15 minutes for submission.

1. Given function $h(x)$ such that

$$h(x) = \begin{cases} 6 - kx, & x \leq 2 \\ x^2 - 10, & 2 < x \leq 4 \\ \dfrac{12}{x-2}, & x > 4 \end{cases}$$

   i. Find k if $\lim_{x \to 2} h(x)$ exists.

(2 marks)

   ii. Determine whether $h(x)$ is continuous at $x = 4$.

(3 marks)

2. Let $f(x) = \dfrac{x-1}{x^2-1}$. Find

$$\lim_{x \to \infty} x f(x) + \lim_{x \to 1} \frac{x}{f(x)}.$$

(4 marks)

3. Use the First Principle to find the derivative of $f(x) = \dfrac{1}{2\sqrt{x}+1}$.

(4 marks)

4. Find $\dfrac{dy}{dx}$ for the curve given as $ye^{x+y} = \sin\left(3x - y^2\right)$ by using implicit differentiation.

(5 marks)

5. Given a set of parametric equations

$$x = \ln(1+t^2) \qquad \text{and} \qquad y = 2t^3 - 2t^2.$$

i. Find $\dfrac{dy}{dx}$ in terms of t.

(3 marks)

ii. Find $\dfrac{d^2y}{dx^2}$ at $t = 1$.

(3 marks)

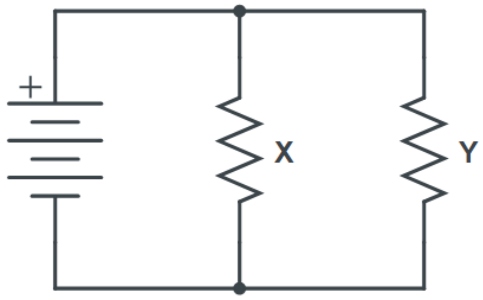

Figure 1: Figure for Question 6

6. Figure 1 shows a circuit where X and Y are the resistors. The total resistance R, is given by

$$\frac{1}{R} = \frac{1}{X} + \frac{1}{Y}.$$

The resistances of X and Y are 30 ohms and 50 ohms respectively, and the resistances increase over time.

i. Write an equation of related rate of change involving $\dfrac{dR}{dt}$, $\dfrac{dX}{dt}$ and $\dfrac{dY}{dt}$.

(3 marks)

ii. If $\dfrac{dX}{dt} = 0.03$ ohm/hour and $\dfrac{dY}{dt} = 0.02$ ohm/hour, find $\dfrac{dR}{dt}$.

(3 marks)

END OF QUESTIONS

1) $$h(x) = \begin{cases} 6 - kx & x < 2 \\ x^2 - 10 & 2 < x \leq 4 \\ \dfrac{12}{x-2} & x > 4 \end{cases}$$

(i) $$\lim_{x \to 2^-} h(x) = \lim_{x \to 2^+} h(x)$$

$$6 - kx = x^2 - 10$$

$$6 - 2k = 4 - 10$$

$$2k = 12$$

$$k = 6$$

(ii) $$\lim_{x \to 4^-} h(x) = x^2 - 10 = 4^2 - 10 = 6$$

$$\lim_{x \to 4^+} h(x) = \frac{12}{x-2} = \frac{12}{4-2} = 6$$

$$\lim_{x \to 4^-} h(x) = \lim_{x \to 4^+} h(x), \quad \lim_{x \to 4} h(x) \text{ exists}$$

$$h(4) = x^2 - 10 = 6$$

$\therefore$ since $\lim_{x \to 4} h(x) = h(4)$, $h(x)$ is continuous at $x = 4$

2) $$f(x) = \frac{x-1}{x^2-1}$$

$$\lim_{x \to \infty} x f(x) = \frac{x(x-1)}{x^2-1} = \frac{x^2 - x}{x^2 - 1} = \frac{\frac{x^2}{x^2} - \frac{x}{x^2}}{\frac{x^2}{x^2} - \frac{1}{x^2}} = \frac{1 - \frac{1}{x}}{1 - \frac{1}{x^2}} = \frac{1-0}{1-0} = 1$$

$$\lim_{x \to 1} \frac{x}{f(x)} = \frac{x}{\frac{x-1}{x^2-1}} = \frac{x(x^2-1)}{x-1} \times \frac{x+1}{x+1} = \frac{x(x^2-1)(x+1)}{x^2-1} = x(x+1)$$
$$= 2$$

$1 + 2 = 3$

3) $$f'(x) = \lim_{h \to 0} f(x+h) = \frac{\frac{1}{2\sqrt{x+h}+1} - \frac{1}{2\sqrt{x}+1}}{h} = \frac{\frac{2\sqrt{x} \cancel{+1} - 2\sqrt{x+h} \cancel{-1}}{(2\sqrt{x+h}+1)(2\sqrt{x}+1)}}{h}$$

$$= \left(\frac{2\sqrt{x} - 2\sqrt{x+h}}{(2\sqrt{x+h}+1)(2\sqrt{x}+1)} \times \frac{2\sqrt{x} + 2\sqrt{x+h}}{2\sqrt{x} + 2\sqrt{x+h}} \right) \left(\frac{1}{h} \right)$$

$$= \left(\frac{4x - 4x - 4\cancel{h}}{2\sqrt{x}(2\sqrt{x+h}+1)(2\sqrt{x}+1) + 2\sqrt{x+h}(2\sqrt{x+h}+1)(2\sqrt{x}+1)} \right) \left(\frac{1}{\cancel{h}} \right)$$

$$= \frac{-4}{2\sqrt{x}(2\sqrt{x}+1)(2\sqrt{x}+1) + 2\sqrt{x}(2\sqrt{x}+1)(2\sqrt{x}+1)}$$

$$= \frac{-4}{2(2\sqrt{x})(2\sqrt{x}+1)^2}$$

$$= \frac{-\cancel{4}}{\cancel{4}\sqrt{x}(2\sqrt{x}+1)^2}$$

$$= \frac{-1}{\sqrt{x}(2\sqrt{x}+1)^2}$$

4) $ye^{x+y} = \sin(3x - y^2)$

$\frac{dy}{dx}(ye^{x+y}) = \frac{dy}{dx}(\sin 3x - y^2)$

$u = y$

$v = e^{x+y}$

$u' = 1\left(\frac{dy}{dx}\right)$

$v' = e^{x+y}\left(1 + \frac{dy}{dx}\right)$

$y = \sin(3x - y^2)$

$u = 3x - y^2$, $\frac{du}{dx} = 3 - 2y\left(\frac{dy}{dx}\right)$

$y = \sin u$

$\frac{dy}{du} = \cos u$

$\frac{dy}{dx} = \cos(3x - y^2)(3 - 2y)\left(\frac{dy}{dx}\right)$

$$\frac{vu' - vu'}{v^2} = \frac{\cancel{e^{x+y}}\left(\frac{dy}{dx}\right) - y\cancel{(e^{x+y})}\left(1 + \frac{dy}{dx}\right)}{(e^{x+y})^{\cancel{2}}} = \cos(3x - y^2)(3 - 2y)\left(\frac{dy}{dx}\right)$$

$$\frac{\left(\frac{dy}{dx}\right) - y - \frac{dy}{dx}}{(e^{x+y})} = \cos(3x - y^2)(3 - 2y)\left(\frac{dy}{dx}\right)$$

$$-y = \cos(3x - y^2)(3 - 2y)\left(\frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = \frac{-y}{\cos(3x - y^2)(3 - 2y)}$$

dy

5) $x = \ln(1 + t^2)$, $y = 2t^3 - 2t^2$

(i) $x = \ln(1 + t^2)$

$x = \ln(u)$ $\quad \frac{dx}{du} = \frac{1}{u}$

$u = 1 + t^2$

$\frac{du}{dt} = 2t$

$\frac{dx}{dt} = \frac{1}{1+t^2}(2t)$

$= \frac{2t}{1+t^2}$

$y = 2t^3 - 2t^2$

$\frac{dy}{dt} = 6t^2 - 4t$

$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

$= 6t^2 - 4t\left(\frac{1+t^2}{2t}\right)$

$= 6t^{\cancel{2}}\left(\frac{1+t^2}{\cancel{2t}}\right) - 4t\left(\frac{1+t^2}{\cancel{2t}}\right)$

$= 3t(1 + t^2) - 2(1 + t^2)$

$= 3t + 3t^3 - 2 - t^2$

$= 3t^3 - t^2 + 3t - 2$

(ii) $\frac{d^2y}{dx^2} = 9t^2 - 2t + 3$, $t = 1$

$= 9(1^2) - 2(1) + 3$

$= 9 - 2 + 3$

$= 10$

$R^{-1} = x^{-1} + Y^{-1}$

$\frac{R^{-1}}{t} = \frac{x^{-1}}{t} + \frac{Y^{-1}}{t}$

6) $X : 30\Omega$, $Y : 50\Omega$

$\frac{1}{R} = \frac{1}{x} + \frac{1}{Y}$

$R = 18.75\Omega$

(i) $\frac{R}{t} = \frac{1}{t(\frac{1}{x}+\frac{1}{y})} = \frac{1}{t(\frac{x+Y}{xY})} = \frac{xy}{tx+ty}$

$u = xy \quad u' = 0$

$v = tx + ty$, $v' = x + y$

$\frac{dR}{dt} = \frac{-xy(x+y)}{(tx+ty)^2}$

$\frac{dx}{dt} =$

$\frac{dy}{dt} =$

(ii) $\frac{dx}{dt} = 0.03$ $\frac{dy}{dt} = 0.02$, if given 1 hour

$x = 0.03\Omega$ $Y = 0.02\Omega$

$\frac{1}{R} = \frac{1}{30.03} + \frac{1}{50.02}$

$R = 18.76\Omega$

$\frac{dR}{dt} = \frac{R - R_0}{t}$

$= 0.01$ ohm/hour

LIST OF FORMULA

Trigonometric
$\cos^2 x + \sin^2 x = 1$
$1 + \tan^2 x = \sec^2 x$
$\cot^2 x + 1 = \operatorname{cosec}^2 x$
$\sin 2x = 2\sin x \cos x$
$\cos 2x = \cos^2 x - \sin^2 x$ $= 2\cos^2 x - 1$ $= 1 - 2\sin^2 x$
$\tan 2x = \dfrac{2\tan x}{1-\tan^2 x}$
$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
$\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
$2\sin x \cos y = \sin(x+y) + \sin(x-y)$
$2\sin x \sin y = -\cos(x+y) + \cos(x-y)$
$2\cos x \cos y = \cos(x+y) + \cos(x-y)$

Logarithm
$a^x = e^{x\ln a}$
$\log_a x = \dfrac{\log_b x}{\log_b a}$

Differentiations
$\dfrac{d}{dx}[k] = 0,\ k$ constant
$\dfrac{d}{dx}[x^n] = nx^{n-1}$
$\dfrac{d}{dx}[e^x] = e^x$
$\dfrac{d}{dx}[\ln\lvert x\rvert] = \dfrac{1}{x}$
$\dfrac{d}{dx}[\cos x] = -\sin x$
$\dfrac{d}{dx}[\sin x] = \cos x$
$\dfrac{d}{dx}[\tan x] = \sec^2 x$
$\dfrac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$
$\dfrac{d}{dx}[\sec x] = \sec x \tan x$
$\dfrac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$
Parametric Differentiation
$\dfrac{dy}{dx} = \dfrac{dy}{dt} \times \dfrac{dt}{dx}$ $\dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right) \times \dfrac{dt}{dx}$